

DINOv3-Large with LoRA Adaptation for Identity Document Fraud Detection

Technical Report — FREUID Challenge 2026 (IJCAI-ECAI)

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Abstract

We present a solution for the FREUID Challenge 2026, which evaluates identity document fraud detection across diverse document types with emphasis on out-of-distribution generalization. Our approach employs DINOv3-Large as a visual backbone with Low-Rank Adaptation (LoRA), enabling efficient fine-tuning while preserving the pre-trained forensic representations. The model is trained using Focal Loss to handle class imbalance, validated with Leave-One-Type-Out (LOTO) cross-validation to simulate the out-of-distribution challenge of the private test set. Our method achieves a public leaderboard FREUID score of 0.04087.

1. Method

1.1 Backbone and Pre-training

We use DINOv3-Large as the visual backbone, loaded via HuggingFace Transformers as a TimmWrapperModel. DINOv3 is a self-supervised Vision Transformer (ViT) pre-trained on large-scale visual data using the DINO objective, producing rich patch-level representations that encode both local forensic artifacts (halftone patterns, JPEG blocking, moiré) and global document structure. These properties make it well-suited for identity document fraud detection across unseen document types.

1.2 Low-Rank Adaptation (LoRA)

Rather than fine-tuning the full 300M+ parameter backbone, we apply LoRA to all attention and MLP layers. This is motivated by recent findings (DINOv3 for Forensics, arXiv:2604.16083) showing that LoRA achieves $F1=0.774$ versus $F1=0.530$ for full fine-tuning on forensic tasks, because frozen pre-trained features encode generalizable forensic information that full fine-tuning destroys on small datasets. LoRA injects trainable rank decomposition matrices A and B in parallel to each frozen weight matrix W :

$$W' = W + (B \times A) \times (\alpha / r)$$

We apply LoRA to all qkv, proj, fc1, and fc2 layers across all transformer blocks (4 layers $\times$ number_of_blocks replacements total), with rank $r=16$, $\alpha=32$, and dropout=0.05. This yields approximately 5% trainable parameters relative to the total backbone size.

1.3 Classification Head

The CLS token output from DINOv3 (dimension 1024) is passed through a single linear layer to produce a binary fraud logit. We deliberately keep the head simple (no MLP, no dropout) to avoid adding additional regularization that could slow convergence.

1.4 Loss Function

We use Focal Loss (Lin et al., 2017) with $\alpha=0.2$ and $\gamma=2.0$ to down-weight easy genuine samples and focus learning on subtle fraud cases. Focal Loss modulates the standard binary cross-entropy by a factor $(1-p_t)^\gamma$:

$$FL(p_t) = -\alpha_t \times (1 - p_t)^\gamma \times \log(p_t)$$

Label smoothing ($\epsilon=0.05$) and Mixup augmentation ($\alpha=0.4$, probability=0.3) are applied during training as additional regularization against overfitting.

2. Training Protocol

2.1 Validation Strategy

We use Leave-One-Type-Out (LOTO) cross-validation, which removes one complete document type from training and uses it as the validation set. This directly simulates the private test evaluation scenario where the model encounters two previously unseen document types. With 5 document types in the training data (BENIN/DL, EGYPT/DL, GUINEA/DL, MAURITIUS/ID, MOZAMBIQUE/DL), LOTO produces 5 folds, each validating on one held-out type.

2.2 Data Augmentation

Augmentations are carefully selected to simulate the print-and-capture pipeline dominant in the private test set:

- JPEG compression (quality 40-90, $p=0.5$) — simulates codec artifacts after recapture
- Motion/Gaussian blur (kernel 5, $p=0.4$) — simulates camera shake and scanner blur
- Random brightness/contrast ($\pm 25\%$, $p=0.5$) — simulates lighting variation
- Horizontal flip ($p=0.5$) — invariance to document orientation
- Perspective transform (scale 0.02-0.05, $p=0.3$) — simulates off-angle capture

2.3 Hyperparameters

Hyperparameter	Value	Rationale
Backbone	DINOv3-Large	Best forensic generalization per literature
Image size	448 × 448 px	Balance detail vs. memory
LoRA rank r	16	Sufficient capacity, low overfitting risk
LoRA α	32	Standard 2× scaling
Batch size	16 (effective 16)	Fit in 95GB VRAM
Learning rate	2e-5	Stable convergence with LoRA
Epochs	20 (early stop pat=4)	Prevent overfitting
Optimizer	AdamW	Weight decay 0.01
Scheduler	OneCycleLR	Cosine annealing with warmup
AMP dtype	bfloat16	Blackwell GPU native support

3. Results

3.1 Public Leaderboard

Metric	Score
Public FREUID Score	0.04087
Validation Strategy	LOTO (5 folds)
Ensemble	5 checkpoints × 3 TTA

3.2 Test-Time Augmentation

At inference time, we apply 3-pass TTA: (1) original image, (2) horizontal flip, (3) brightness/contrast variation. Predictions from all 5 LOTO checkpoints and 3 TTA passes are averaged (15 total inferences per image) to produce the final fraud score.

4. Reproducibility

4.1 Environment

Component	Version / Details
Python	3.12
PyTorch	2.5.1 + CUDA 12.1
Transformers	4.45.0
timm	latest
albumentations	latest
GPU	NVIDIA RTX PRO 6000 Blackwell Server Edition (95GB VRAM)
OS	Ubuntu 24 (Kaggle environment)

4.2 Training Procedure

The full training pipeline uses Leave-One-Type-Out cross-validation producing 5 checkpoints. Training was conducted on Kaggle with the RTX PRO 6000 Blackwell (95GB VRAM). Each fold trains for up to 20 epochs with early stopping (patience=4). The total training time is approximately 8-10 hours for all 5 folds. Random seed is fixed at 342 for full reproducibility.

4.3 Dependencies

```
torch==2.5.1 transformers==4.45.0 timm albumentations scikit-learn pandas
pillow tqdm numpy
```

5. References

- [1] DFF-Adapter (AAAI 2025): Fine-Grained DINO Tuning with Dual Supervision for Face Forgery Detection. arXiv:2511.12107
- [2] DINOv3 for Forensics (2026): DINOv3 Beats Specialized Detectors. arXiv:2604.16083
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- [4] PAD-IDCard 2024 (IJCB): First Competition on Presentation Attack Detection on ID Card. arXiv:2409.00372
- [5] Lin et al. (2017): Focal Loss for Dense Object Detection. ICCV 2017.
- [6] Hu et al. (2022): LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
- [7] Oquab et al. (2023): DINOv2: Learning Robust Visual Features without Supervision. TMLR 2024.